

RESEARCH METHODOLOGY, IMPLEMENTATION & VALIDATION PLAN

AI-Based Respiratory Sound Screening
Noise-Aware Classification & Explainability Validation

Research type	Applied experimental research with a design-science (build-and-evaluate) component
Core research focus	Joint robustness + explainability evaluation under controlled noise
Primary evaluation axis	Classification robustness, explanation stability, explanation faithfulness, and joint reliability

Working reference document derived from the uploaded research methodology framework.

Document Roadmap

1. Research Paper Methodology: Section-by-Section Workup and Flow
2. Implementation Plan, Feature Workup & Modern Tooling
3. Mathematics, Graphical Studies & Validation Protocol
4. End-to-End Experimental & Prototype Flow

Central principle: the paper should not stop at a basic “train a CNN, predict” pipeline. Every major claim is connected to a measurable experiment: controlled SNR degradation, calibration, explanation stability, explanation faithfulness, a joint reliability measure, ablations, statistical testing, and external validation.

PART 1 — Research Paper Methodology

1.1 Overall paper logic

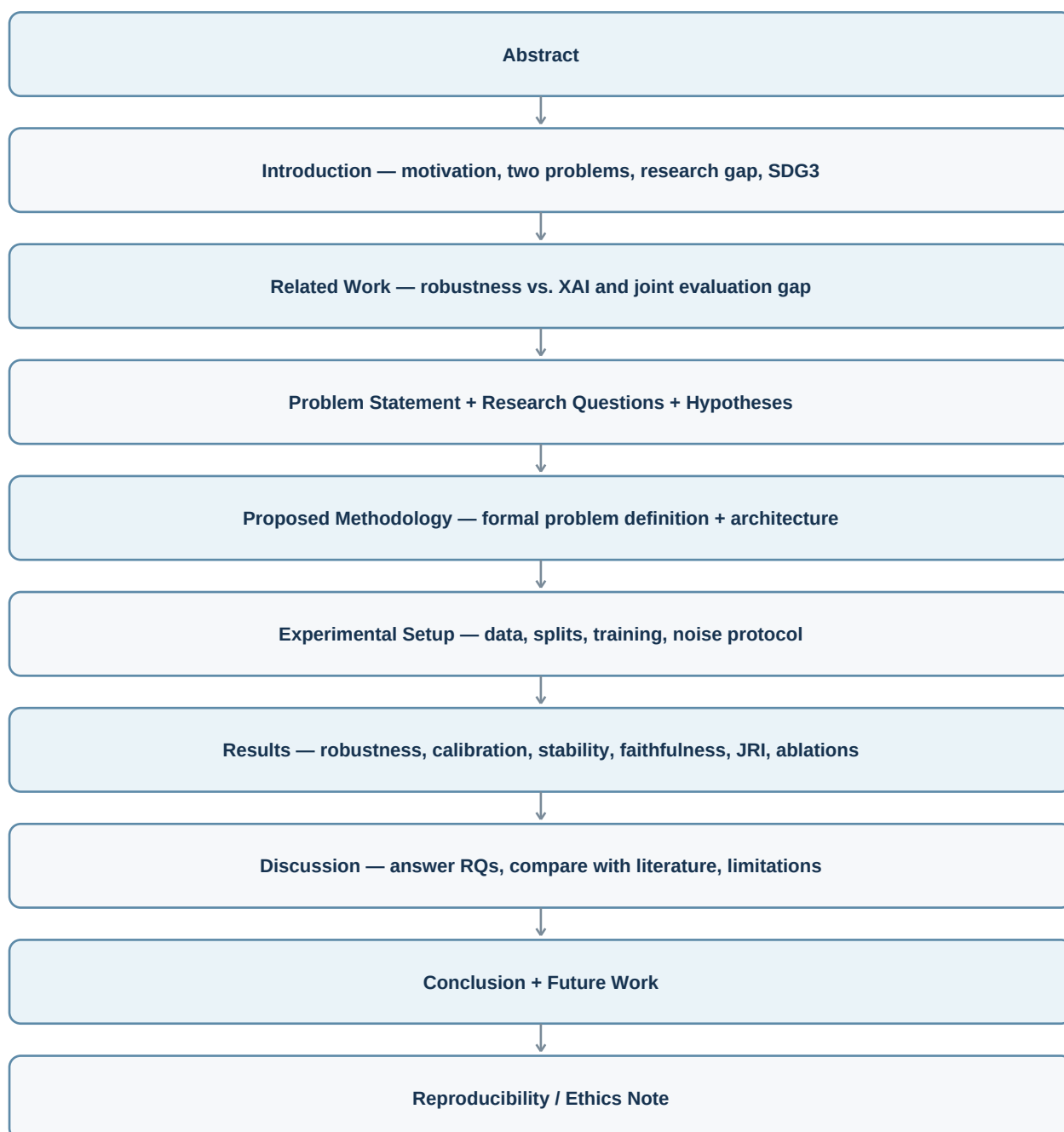


Figure 1. Recommended research-paper dependency flow.

1.2 Research type

This is applied experimental research with a design-science (build-and-evaluate) component: the team builds a pipeline and empirically evaluates it under systematically varied noise conditions.

1.3 Section-by-section workup

Section	What goes in it	When to write
Abstract	Problem → gap → method → quantitative result → contribution	Last
Introduction	Context, noise + explainability problems, gap, SDG3, contributions, roadmap	Early; revise after results
Related Work	Study/task/robustness/XAI/joint-evaluation comparison	Early
Problem + RQs	2–3 formal research questions and hypotheses	Early
Methodology	Formal definition, architecture, seven stages, design choices	Mid-project

Section	What goes in it	When to write
Experimental Setup	Dataset, splits, preprocessing, hyperparameters, noise protocol, reproducibility	Before final runs
Results	Clean baseline, robustness, calibration, XAI stability/faithfulness, JRI, ablations, statistics	Last
Discussion	Answer RQs, position against literature, limitations, threats to validity	After results
Conclusion	Summary + future work mapped to enhancements	Last
Ethics / Reproducibility	Dataset license, not-a-diagnostic-tool statement, code/data availability	Last

1.4 Formal Research Questions & Hypotheses

RQ1 — Robustness

Does noise-aware preprocessing significantly reduce classification-performance degradation caused by additive environmental/device noise across a controlled SNR sweep?

H1: F1 degradation ($\Delta F1 = F1_{\text{clean}} - F1_{\text{SNR}}$) is significantly smaller for the noise-aware pipeline.

RQ2 — Explanation stability

Do saliency-based explanations for the same clip remain spatially consistent as noise increases?

H2: Heatmap similarity decreases with decreasing SNR, and the decrease is smaller for the noise-aware pipeline.

RQ3 — Joint reliability

Can robustness and explanation reliability be combined into one measurable index that behaves consistently across noise levels?

H3: The Joint Reliability Index is significantly higher for the noise-aware pipeline across the SNR sweep.

PART 2 — Implementation Plan, Feature Workup & Modern Tooling

2.1 System implementation architecture

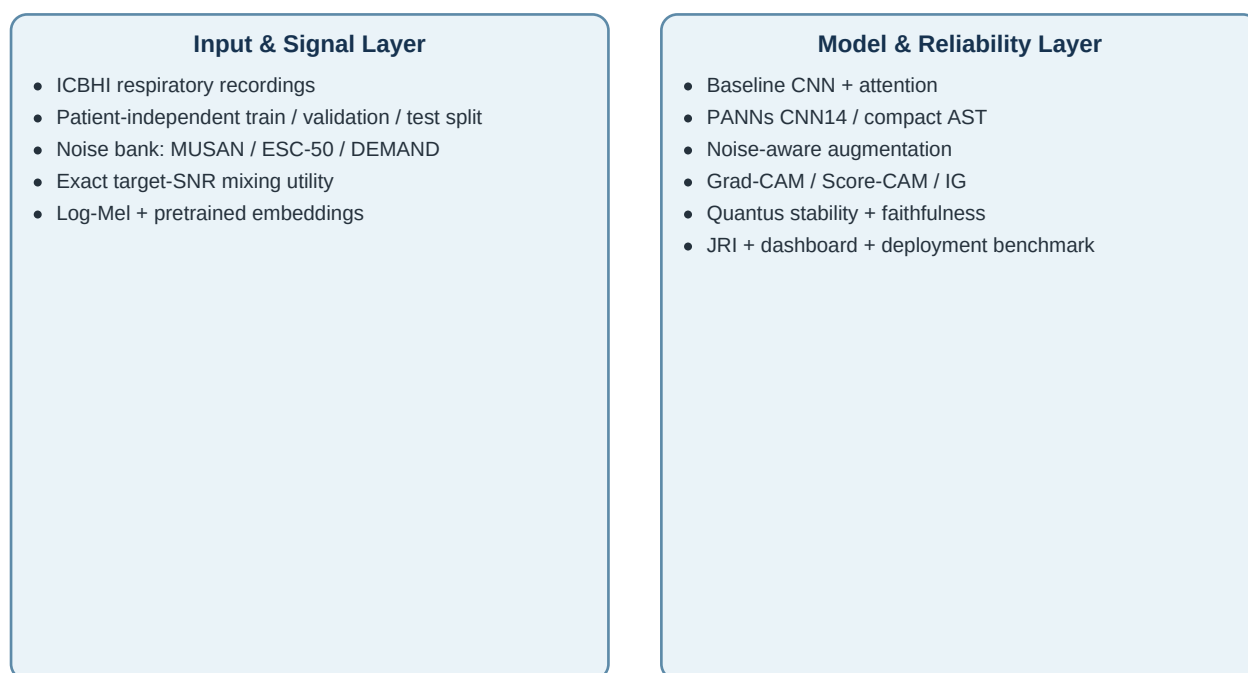


Figure 2. High-level implementation architecture derived from the framework.

2.2 Recommended technology stack

Component	Recommended	Purpose / rationale
Audio I/O / DSP	torchaudio, librosa, soundfile	Audio loading, DSP and spectrogram processing
Primary dataset	ICBHI 2017 Respiratory Sound Database	Standard dataset with patient IDs for patient-independent splitting
Secondary dataset	HF_Lung_V1 or second public corpus	External / cross-corpus validation
Noise	MUSAN, ESC-50, DEMAND + real ambient clips if obtainable	Synthetic + real-world robustness testing
Features	Log-Mel + PANNs embeddings	Strong representation with transfer learning
Baseline	CNN + attention	Controlled from-scratch reference
Upgraded model	PANNs CNN14 or compact Audio Spectrogram Transformer	Modern transfer-learning lever
Robust training	SpecAugment + mixup + randomized-SNR noise	Noise-aware training
Explainability	Captum + Grad-CAM / Score-CAM / Integrated Gradients	Multiple explanation methods
XAI evaluation	Quantus	Quantitative faithfulness / robustness / localization metrics
Tracking	Weights & Biases or MLflow	Track SNR × model × XAI experiment grid
Config / reproducibility	Hydra / YAML + DVC or manifests	Repeatable experiments and data versions
Dashboard	Streamlit + Plotly	Interactive reliability and SNR exploration
Inference	ONNX + onnxruntime	CPU/resource-limited deployment
Testing / statistics	pytest + scipy + statsmodels + sklearn.metrics	Prototype tests and statistical analysis

2.3 Feature / engineering workup — actual build order

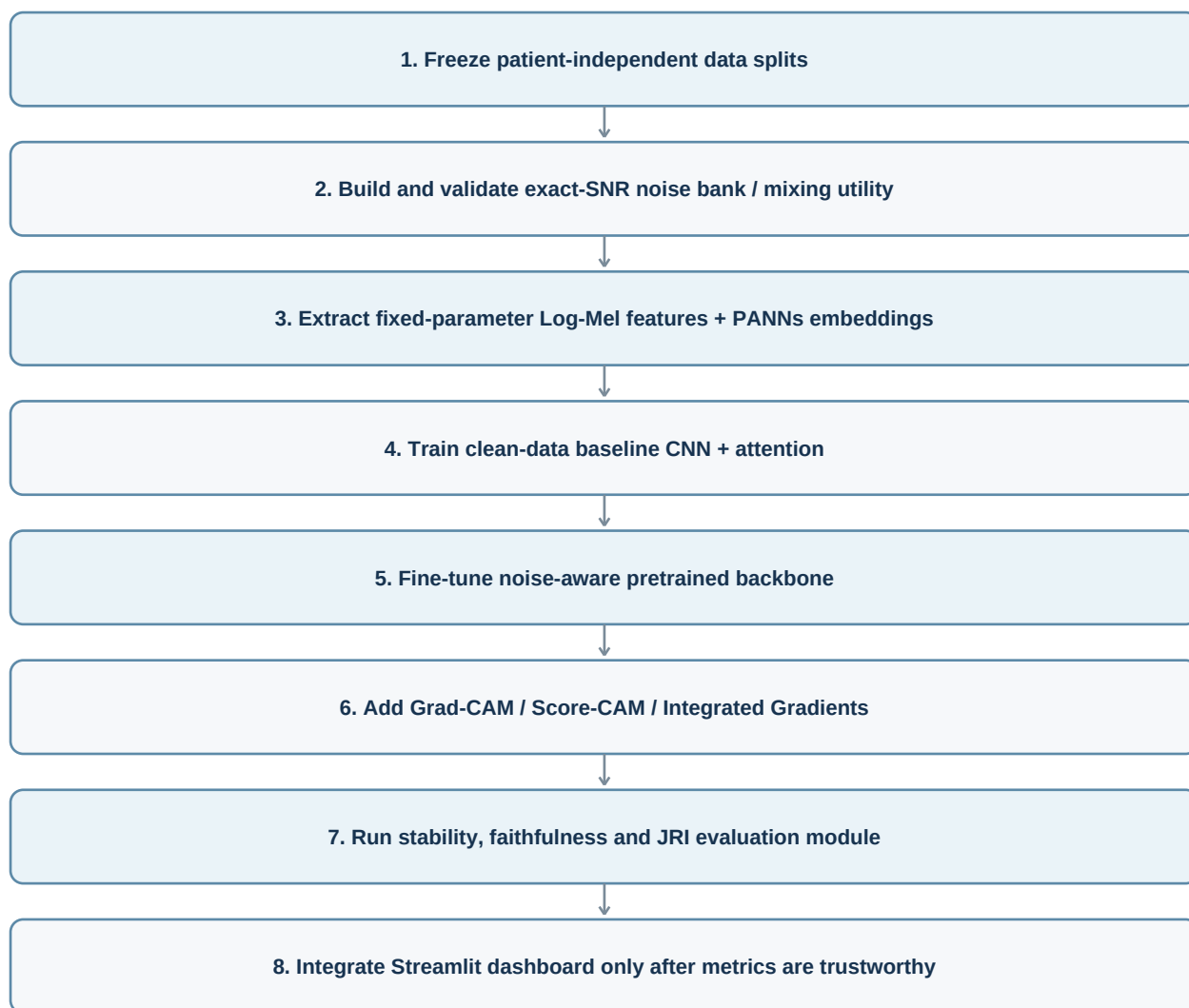


Figure 3. Recommended build order. The dashboard is deliberately last.

2.4 Ten-week phased plan

Phase	Weeks	Focus	Suggested ownership
0	1	Data audit, patient-independent split, noise bank	Shared
1	2–3	Feature extraction + baseline CNN	Member A
2	3–5	Pretrained fine-tuning + noise-aware augmentation	A + B
3	4–6	Explainability wrappers	Member B
4	5–7	Reliability metrics + Quantus	Member C
5	7–8	SNR sweep, statistics, ablations	Shared
6	8–9	Dashboard integration	Member C
7	9–10	Results → Discussion → Abstract	Shared

PART 3 — Mathematics, Graphical Studies & Validation Protocol

3.1 Signal-processing formalism

Short-Time Fourier Transform:

$$X(t, f) = \sum_n x[n] \cdot w[n-tH] \cdot e^{(-j2\pi fn/N)}$$

Mel filterbank energy:

$$M[t, m] = \sum_f |X[t, f]|^2 \cdot H_m(f)$$

Log-Mel feature:

$$S[t, m] = \log(M[t, m] + \epsilon)$$

3.2 Controlled noise injection

Given a clean signal x and noise n , scale the noise to an exact target SNR so that every experiment uses the same controlled perturbation protocol.

$$\begin{aligned} \text{SNR_dB} &= 10 \cdot \log_{10}(P_{\text{signal}} / P_{\text{noise}}), \text{ where } P = \text{mean}(x^2) \\ \text{noise_scaled} &= n \cdot \sqrt{P_x / (P_n \cdot 10^{(\text{SNR_dB}/10)})} \\ x_{\text{noisy}} &= x + \text{noise_scaled} \end{aligned}$$

Suggested test conditions: clean, ≈15 dB mild noise, ≈5 dB moderate noise, and ≈0 dB or below heavy noise.

3.3 Classification objective and metrics

Focal loss:

$$\text{FocalLoss} = - \alpha_c \cdot (1 - p_c)^\gamma \cdot \log(p_c)$$

Metric	Formula / output
Precision	TP / (TP + FP)
Recall / Sensitivity	TP / (TP + FN)
Specificity	TN / (TN + FP)
F1	2 · Precision · Recall / (Precision + Recall)
Balanced Accuracy	(Sensitivity + Specificity) / 2
AUROC / AUPRC	One-vs-rest per class

3.4 Confidence calibration

Expected Calibration Error (ECE): $ECE = \sum_b (|B_b|/N) \cdot |\text{acc}(B_b) - \text{conf}(B_b)|$

Brier Score: $(1/N) \sum_i (p_i - y_i)^2$

Graphical study: reliability diagram at each SNR level — predicted confidence bins vs. observed accuracy.

3.5 Explanation stability

For explanation maps A_{clean} and A_{noisy} of the same underlying clip, compute structural and rank-based similarity:

```
SSIM( $A_{\text{clean}}$ ,  $A_{\text{noisy}}$ )
Spearman  $\rho(\text{flatten}(A_{\text{clean}}), \text{flatten}(A_{\text{noisy}}))$ 
IoU_top-k =  $|\text{top-k}(A_{\text{clean}}) \cap \text{top-k}(A_{\text{noisy}})| / |\text{top-k}(A_{\text{clean}}) \cup \text{top-k}(A_{\text{noisy}})|$ 
```

Stability(SNR) = mean similarity across the test set at that SNR, reported for both models.

3.6 Explanation faithfulness

Use insertion/deletion AUC and AOPC to test whether highly attributed time-frequency regions actually influence the model's decision. Quantus can be used to standardize this evaluation and add metrics such as Faithfulness Correlation and Pixel-Flipping.

$$\text{AOPC} = (1/K) \sum [f(x) - f(x_{\text{perturbed}})]$$

3.7 Joint Reliability Index (JRI)

The framework proposes combining robustness and explanation stability into one measurable construct:

```
R(SNR) = F1(SNR) / F1(clean)
S(SNR) = Stability(SNR)
JRI(SNR) = 2 * R(SNR) * S(SNR) / (R(SNR) + S(SNR))
AURC = (1/N)  $\sum$  JRI(SNR)
```

Interpretation: the harmonic mean prevents a model that is robust but has collapsing explanations (or vice versa) from receiving a high combined score.

3.8 Statistical validation

Test	Purpose
Wilcoxon signed-rank	Paired baseline vs. proposed JRI/F1 across identical test clips at each SNR
McNemar's test	Paired comparison of correct/incorrect decisions on identical clips
Bootstrap 95% CI	Confidence intervals for F1, stability, JRI and other headline metrics
Friedman + Nemenyi	Comparison across more than two SNR levels
Cohen's d	Effect size for robustness and stability gaps

3.9 Required graphical studies

Figure	X-axis	Y-axis	Purpose
Robustness curve	SNR (dB)	F1 / Balanced Accuracy	Core robustness claim
Reliability diagram	Confidence bins	Observed accuracy	Calibration
Stability curve	SNR (dB)	SSIM / IoU	Explanation stability
JRI curve	SNR (dB)	JRI	Combined reliability result
Confusion matrices	—	—	Class errors by SNR
ROC / PR curves	FPR / Recall	TPR / Precision	Per-class clean vs. noisy
Heatmap gallery	—	—	Clean/noisy explanation comparisons
Stability by class	Class	Stability score	Fragile classes
Ablation chart	Removed component	F1 / JRI	Component contribution
Embedding drift	—	—	Noise movement in representation space
Latency / throughput	Model variant	ms / inference	Deployment feasibility

PART 4 — End-to-End Experimental & Prototype Flow

4.1 Research experiment flow

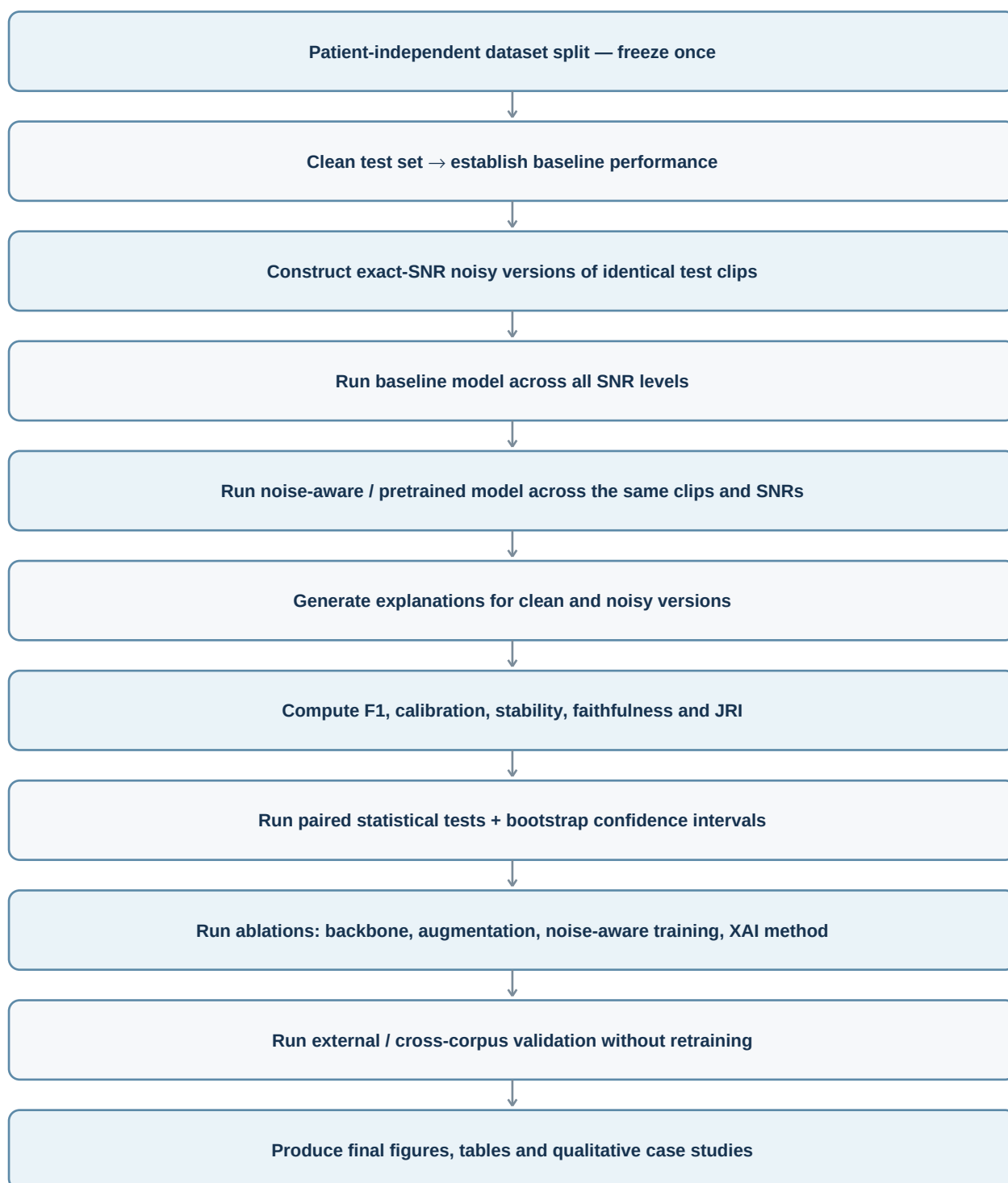


Figure 4. Full research-validation experiment sequence.

4.2 Paper validation protocol

- Patient-independent 5-fold cross-validation; report mean $\pm$ standard deviation.
- Re-implement 1–2 simplified literature baselines on the same data split for apples-to-apples comparison.
- External / cross-corpus validation on a second public dataset without retraining.
- Ablation study: pretrained backbone vs. from-scratch; SpecAugment/mixup on/off; noise-aware training on/off; XAI-method comparison.
- Report statistical tests alongside headline point estimates.

4.3 Prototype / dashboard validation

- Unit tests for quality checking, denoising, feature extraction, inference and explanation-map shapes.
- End-to-end integration test on a held-out clip, including silence, clipped audio and multi-noise-source cases.
- CPU latency / throughput and memory-footprint benchmark.
- Reproducibility check using fixed seeds, pinned dependency versions and a clean checkout.
- Lightweight usability / heuristic review with a mentor or domain-adjacent reviewer; explicitly not clinical validation.

4.4 Final evidence chain

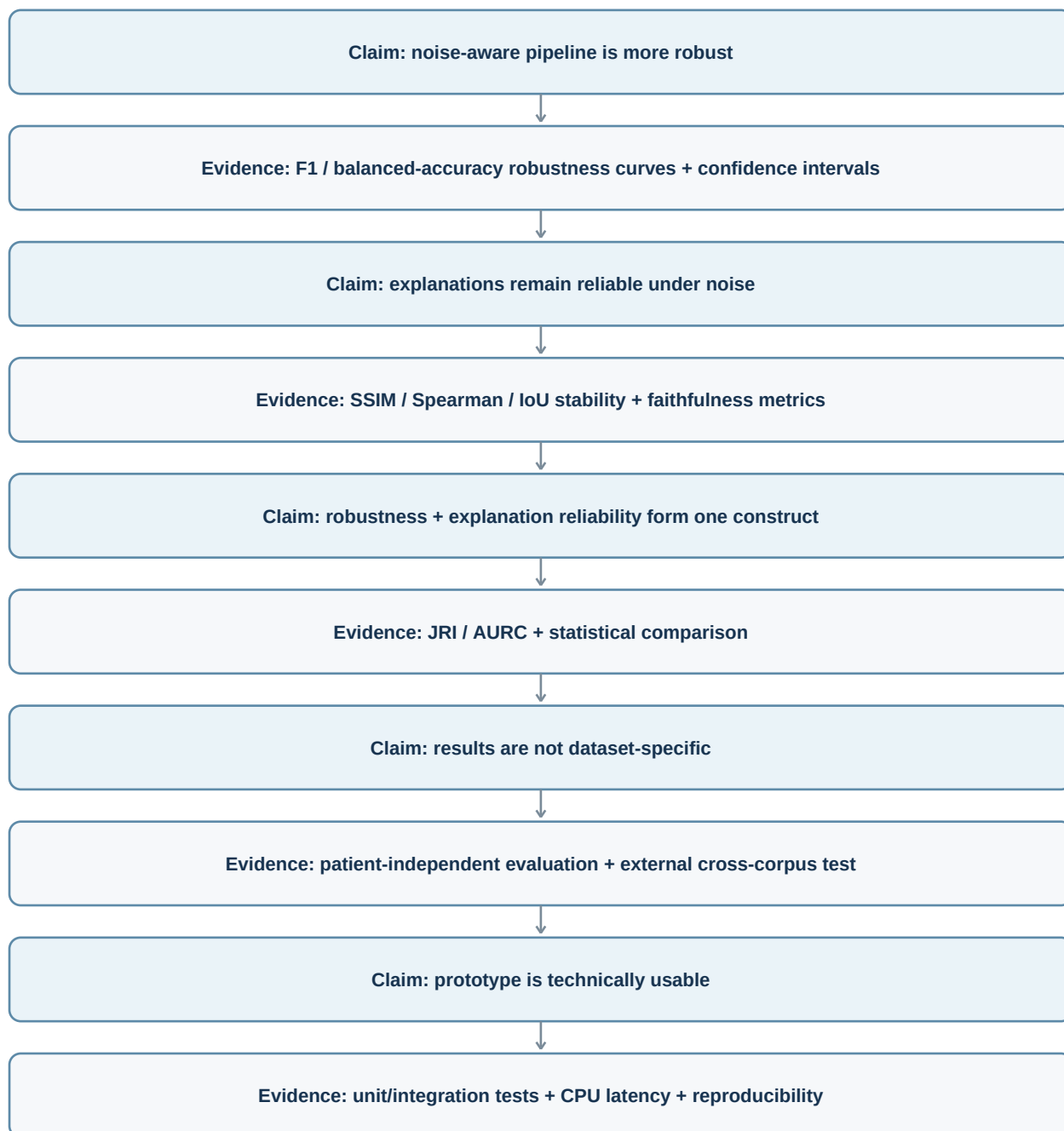


Figure 5. Claim → evidence chain. Every major conclusion should have a corresponding measurable artifact.

Final Research Checklist

- Patient-independent split frozen before model training
- Clean baseline trained and documented
- Noise bank and exact-SNR mixer validated
- Noise-aware model trained with controlled augmentation
- At least two XAI methods implemented
- Stability metrics computed across the SNR sweep
- Faithfulness metrics computed
- Calibration evaluated
- JRI and AURC computed
- Ablation experiments completed
- Statistical significance + confidence intervals reported
- External / cross-corpus validation attempted
- Required plots and qualitative heatmaps generated
- Prototype unit + integration tests passed
- CPU latency / memory benchmark completed
- Reproducibility check completed
- Ethics / non-diagnostic limitation clearly stated

Source basis: the uploaded research methodology framework. The terminology, research questions, implementation sequence, metrics, JRI construction, statistical tests, figures and validation protocols above are formatted from that source rather than replaced with a generic project plan.